

Learning from Demonstration

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UNIVERSITÀ
POLITECNICA
DELLE MARCHE

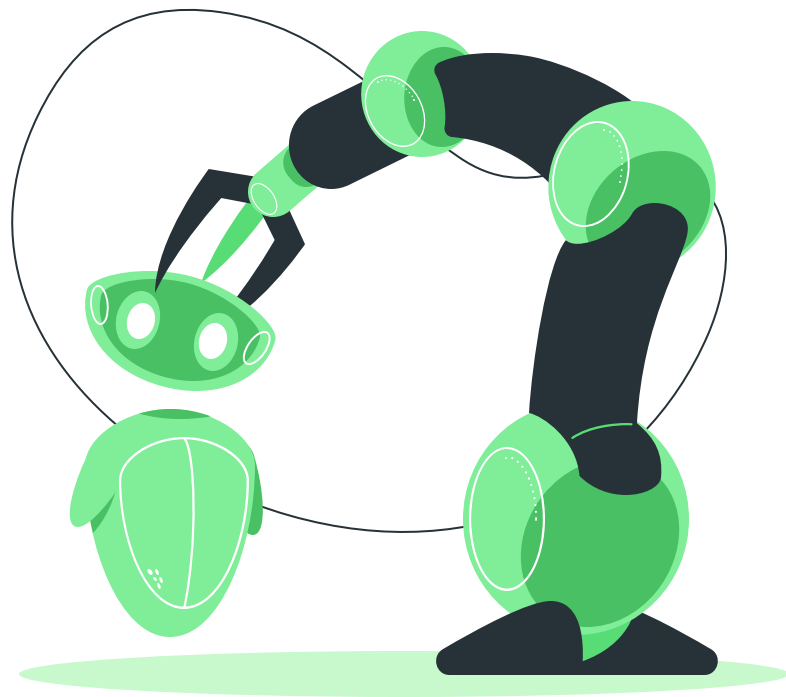


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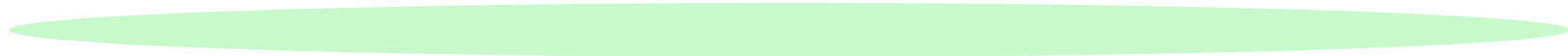
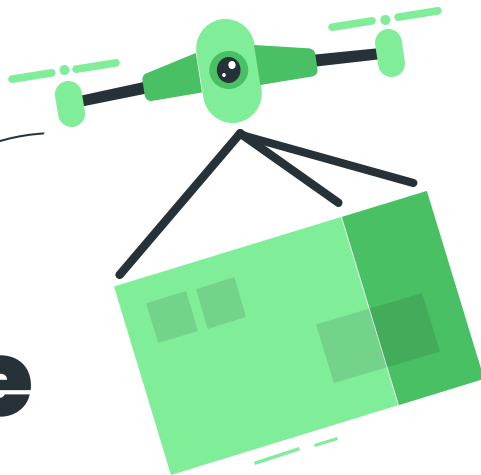
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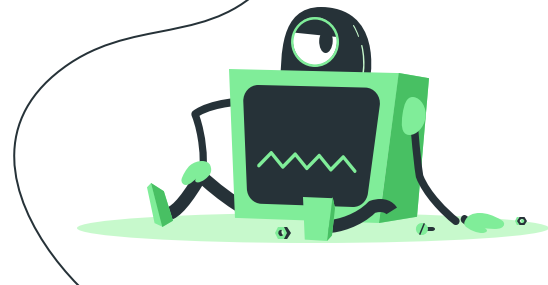
Introduzione

Contestualizziamo.



Contesto

- Robotica
- Industria 4.0
- Human-Robot Collaboration
- Supervised Learning



Definizione

La **Learning from Demonstration** è un paradigma di machine learning nel quale un agente, detto **learner**, è allenato su dati forniti tramite dimostrazioni di task, eseguite da un soggetto denominato **teacher o demonstrator**.



Motivazioni

- **Complessità della programmazione dei robot tradizionali**
- **Approcci di programmazione piu' rapidi ed efficienti**
- **Cobots sempre piu accessibili**
- **Mass Customization**

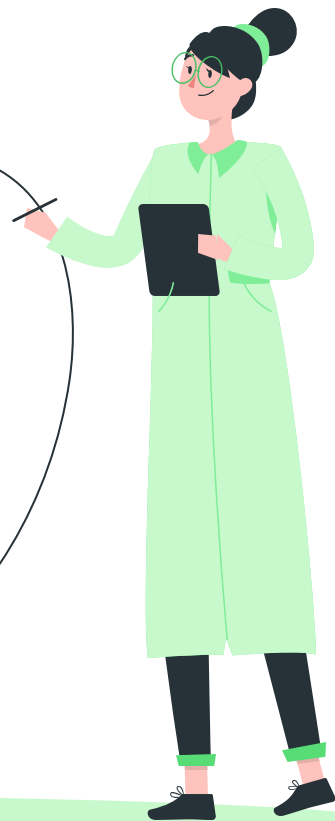




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Workflow

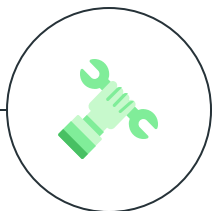
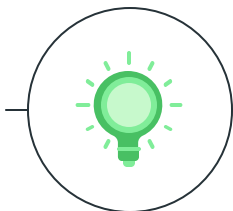
Uno step per domarli
Uno step per ghermirli
Uno step per testarli e nel task migliorarli



Workflow LfD

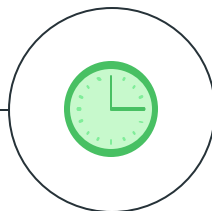
Data Acquisition

Dimostrazione Task e
Raccolta Dati



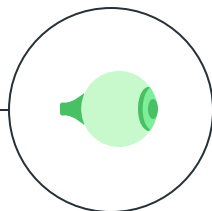
Data Modelling

Derivazione delle policy
di comportamento



Task Execution

Testing policy derivate
Deploy del learner



Learning Refinement

Ottimizzazione policy
tramite esperienza

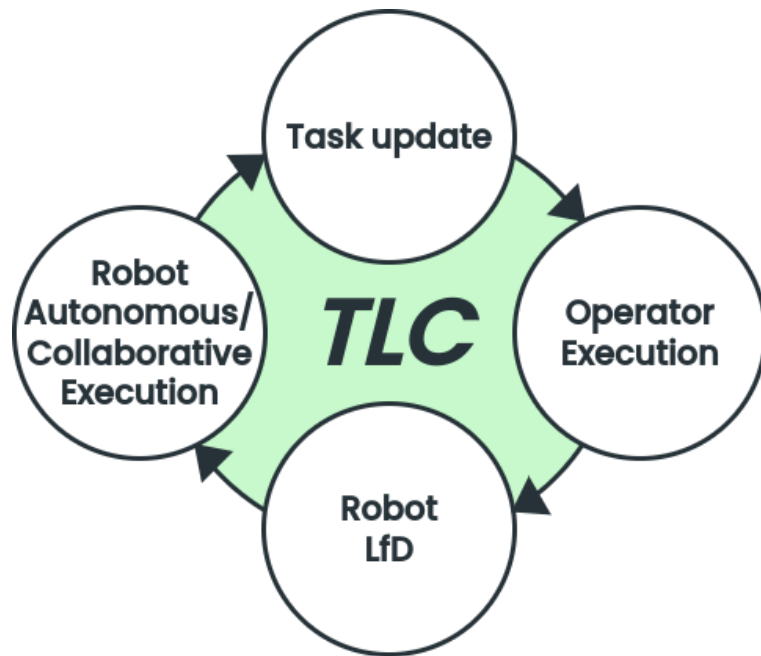
TLC

Teaching

Learning

Collaboration

Il meglio dei due mondi



An illustration featuring a green robot on the left and a person on the right. The robot is standing with its arms raised, and the person is sitting cross-legged, working on a laptop. A thin black line connects the robot's arm to the person's laptop, looping around the central text. The robot has a heart symbol on its chest. The person is wearing glasses and a green shirt. The background is white with a light green oval at the bottom.

3

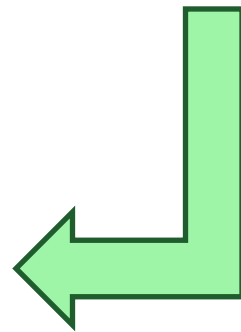
Data Acquisition

Come acquisiamo i dati?

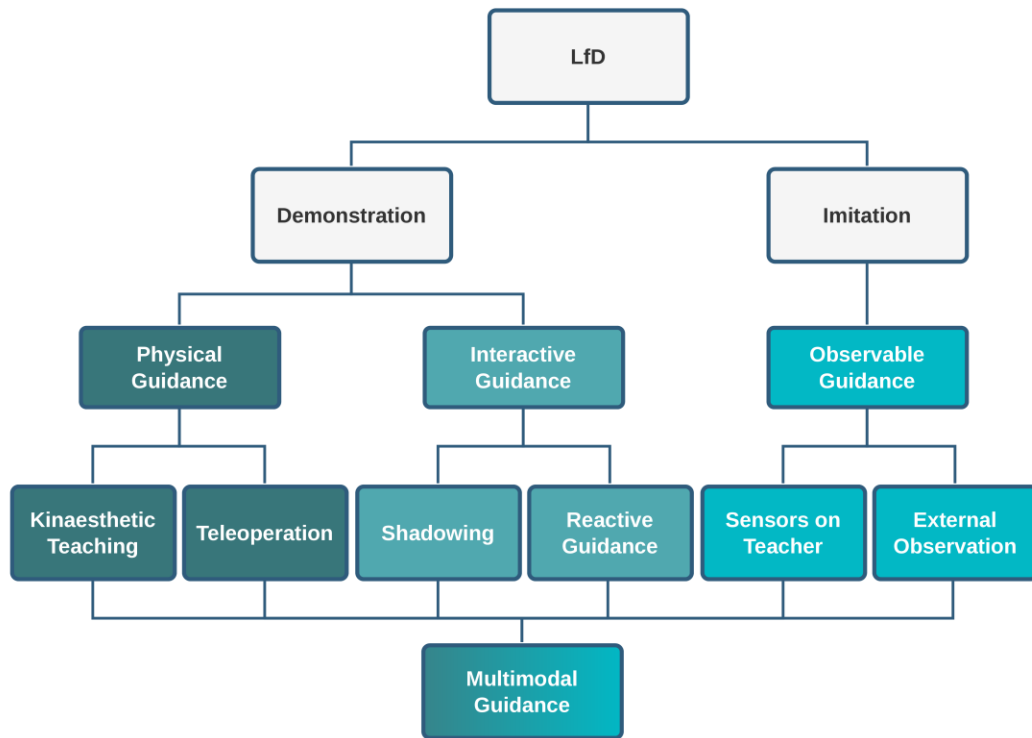
Corrispondenza



	Demonstration	Imitation
	$g_E(z, a) = I(z, a)$	$g_E(z, a) \neq I(z, a)$
$g_R(z, a) = I(z, a)$	Physical Guidance	Sensors on Teacher
$g_R(z, a) \neq I(z, a)$	Interactive Guidance	External Observation



Categorizzazione

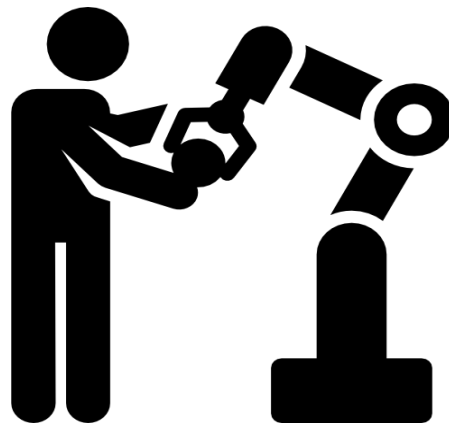


Kinesthetic Teaching

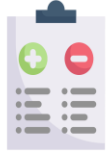


Pros & Cons

- Mapping non necessario
- Richiede poche risorse
- Tecnica intuitiva
- Formazione del teacher
- Problemi con alti DoF
- Co-località stringente

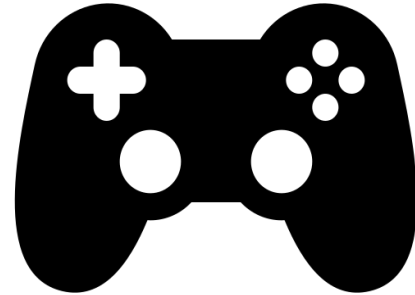


Teleoperation



Pros & Cons

- **Applicabile da remoto**
- **Abilita simulazione**
- **Applicabile su larga scala**
- **Formazione del teacher**
- **Interfaccia complessa**
- **Costoso**
- **Mapping necessario**

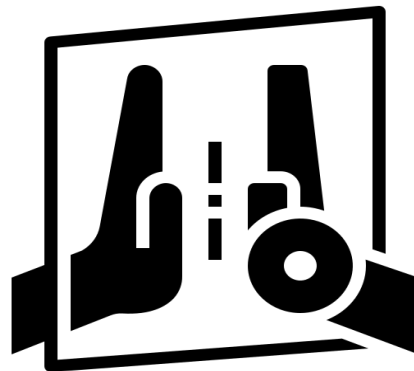


Shadowing



Pros & Cons

- **Intuitivo per il teacher**
- **Correzione real-time**
- **Algoritmi complessi**
- **Costoso**
- **Mapping necessario**

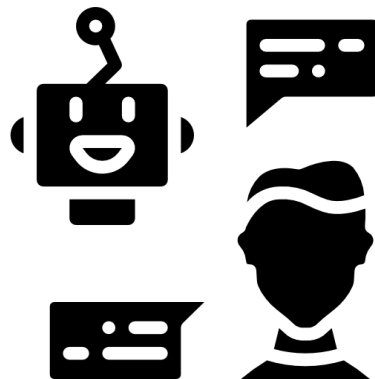


Reactive Guidance

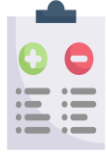


Pros & Cons

- **Diverse metodologie per il feedback**
- **Correzione real-time**
- **Intuitivo**
- **Necessarie interazioni prolungate**
- **Soluzioni sub-ottimali**
- **Rischio overfitting**



Sensor on Teacher



Pros & Cons

- **Dati raccolti accurati**
- **Informazioni di dettaglio**
- **Strumentazione ad-hoc**
- **Costoso**
- **Impedimenti per il teacher**

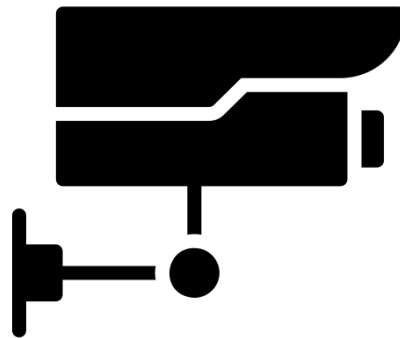


External Observation



Pros & Cons

- **Informazioni sull'ambiente**
- **Riutilizzo sistema d'acquisizione**
- **Costoso**
- **Algoritmi complessi**
- **Dati non accurati**



4

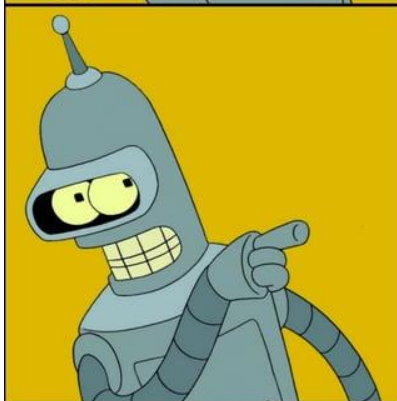
Conclusioni

Quindi quale tecnica utilizziamo?





**DOBBIAMO
SCEGLIERE TRA
6 TECNICHE???**



**POSSIAMO
SCEGLIERE TRA
6 TECNICHE!!!**



Grazie!

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